Non-linear simultaneous equations

1

a $x = 0, y = 0$ and $x = 4, y = 16$

b $x = 0, y = 0$ and $x = -4, y = -8$

c $x = 2, y = -2$ and $x = -3, y = -7$

d $x = 0, y = -16$ and $x = 1, y = -15$

e $x = 3, y = 1$ and $x = 5, y = -15$

f $x = 0, y = 5$ and $x = -2, y = 1$

g $x = 1, y = 3$ and $x = -2, y = 6$

h $x = 4, y = 4$ and $x = -1, y = 9$

i $x = -1, y = 4$ and $x = 11, y = 16$

j $x = 3, y = 4$

k $x = -5, y = -28$ and $x = \frac{3}{2}, y = -2$

l $x = -2, y = 2$ and $x = 1, y = 8$

2

a Example answer: $x \approx 2.3, y \approx 1.3$ and $x \approx -1.3, y \approx -2.3$

b Example answer: $x \approx 1.5, y \approx 5.5$ and $x \approx -2.5, y \approx 1.4$

Applications

3

a $x + y = 60, y = x^2 + 4$

b $(7, 53), (-8, 68)$

4

a $x = -1, x = 3$

b $x = 3$ since the supply needs to be larger than zero.

c $y = 18$

d The equilibrium quantity occurs when 3 thousand units are being produced, which has a corresponding price of 18 dollars per unit.

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